Arduino
Introdução

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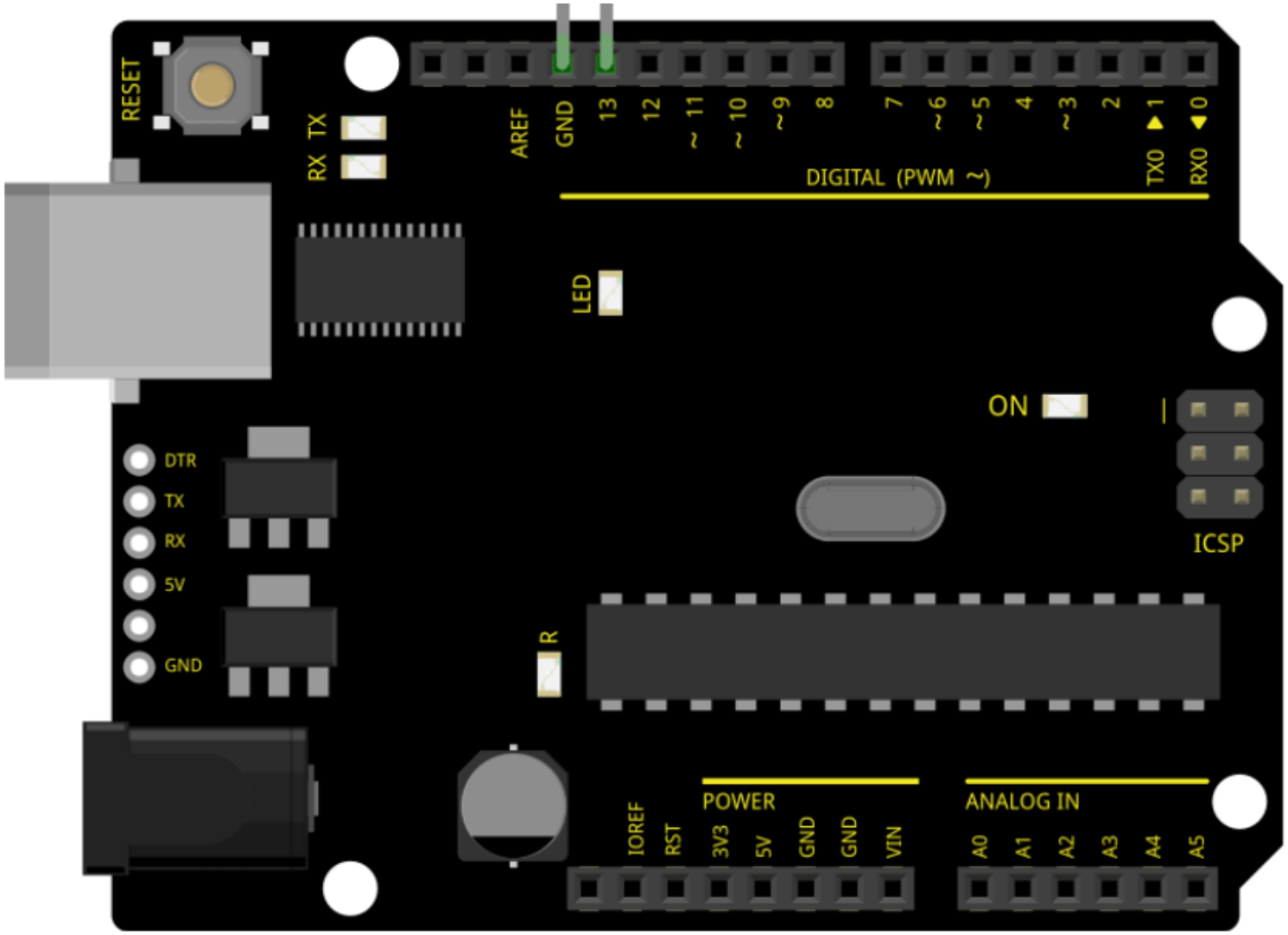
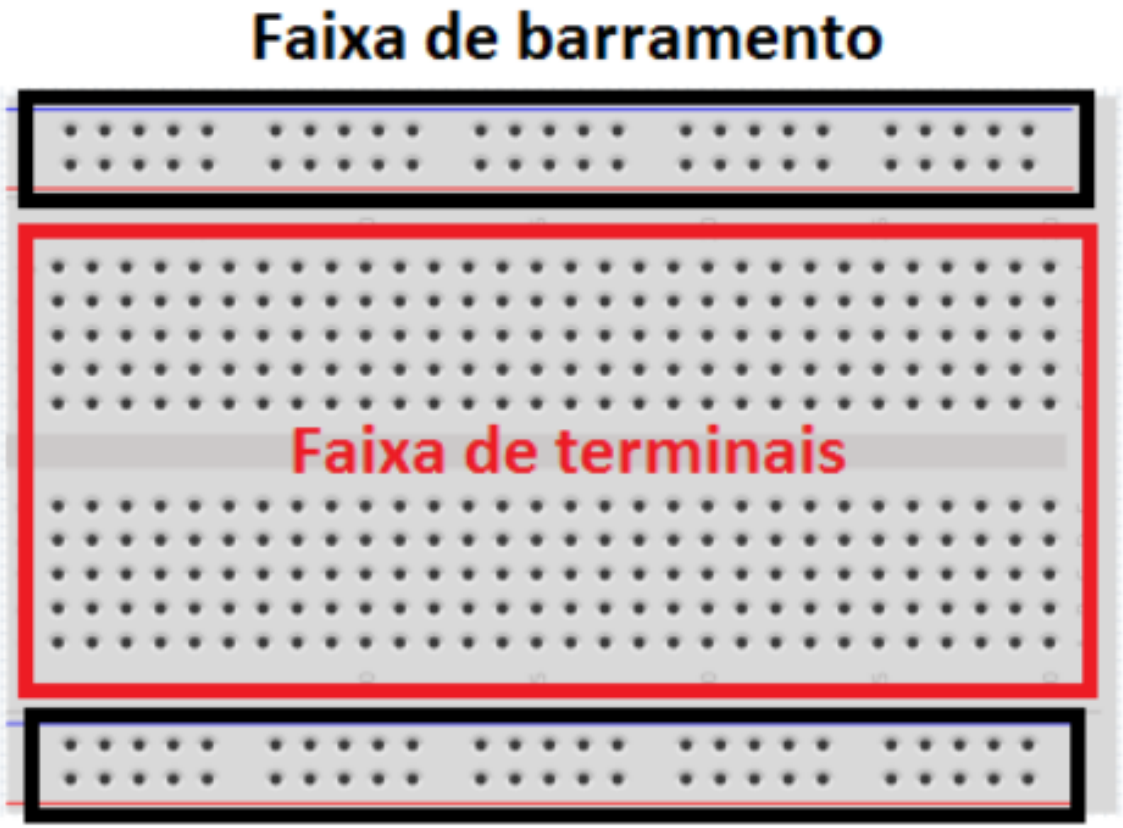
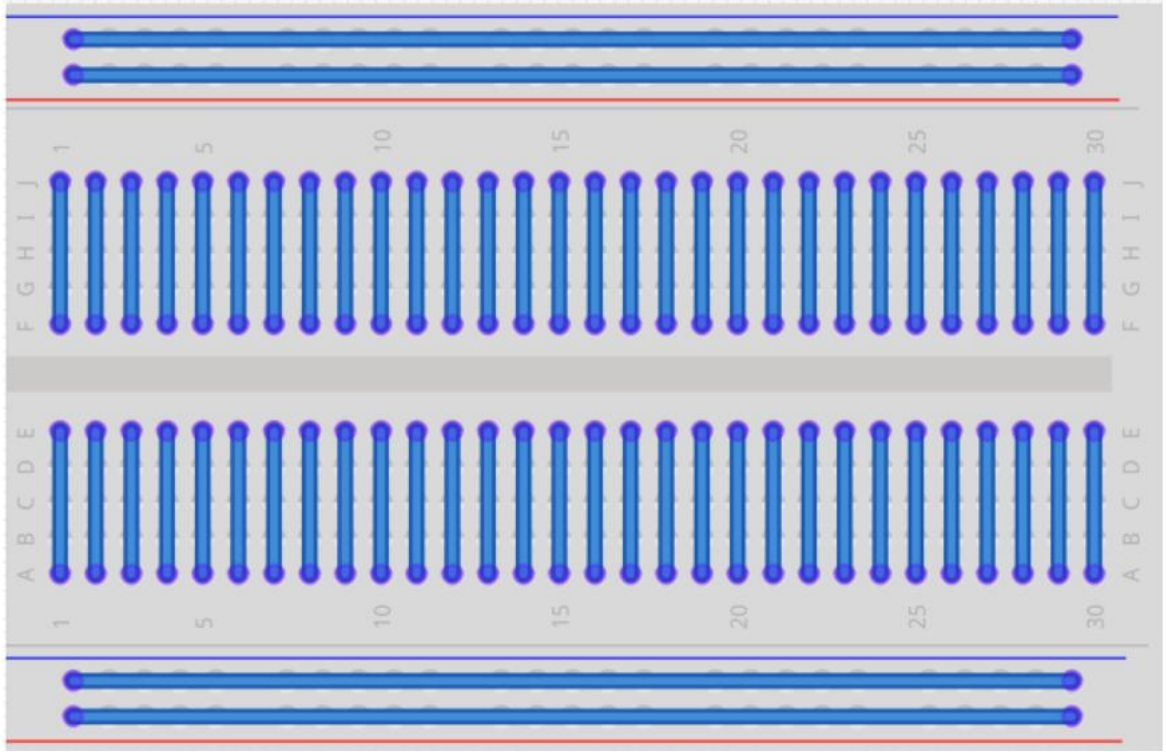
Arduino UNO


RESET
RX
TX
AREF
GND
13
12
~ 11
~ 10
~ 9
8
7
~ 6
~ 5
4
~ 3
2
TX0 ▶ 1
RX0 ◀ 0
DIGITAL (PWM ~)
LED
ON
ICSP
DTR
TX
RX
5V
GND
R
IOREF
RST
POWER
3V3
5V
GND
GND
VIN
ANALOG IN
A0
A1
A2
A3
A4
A5

Protoboard

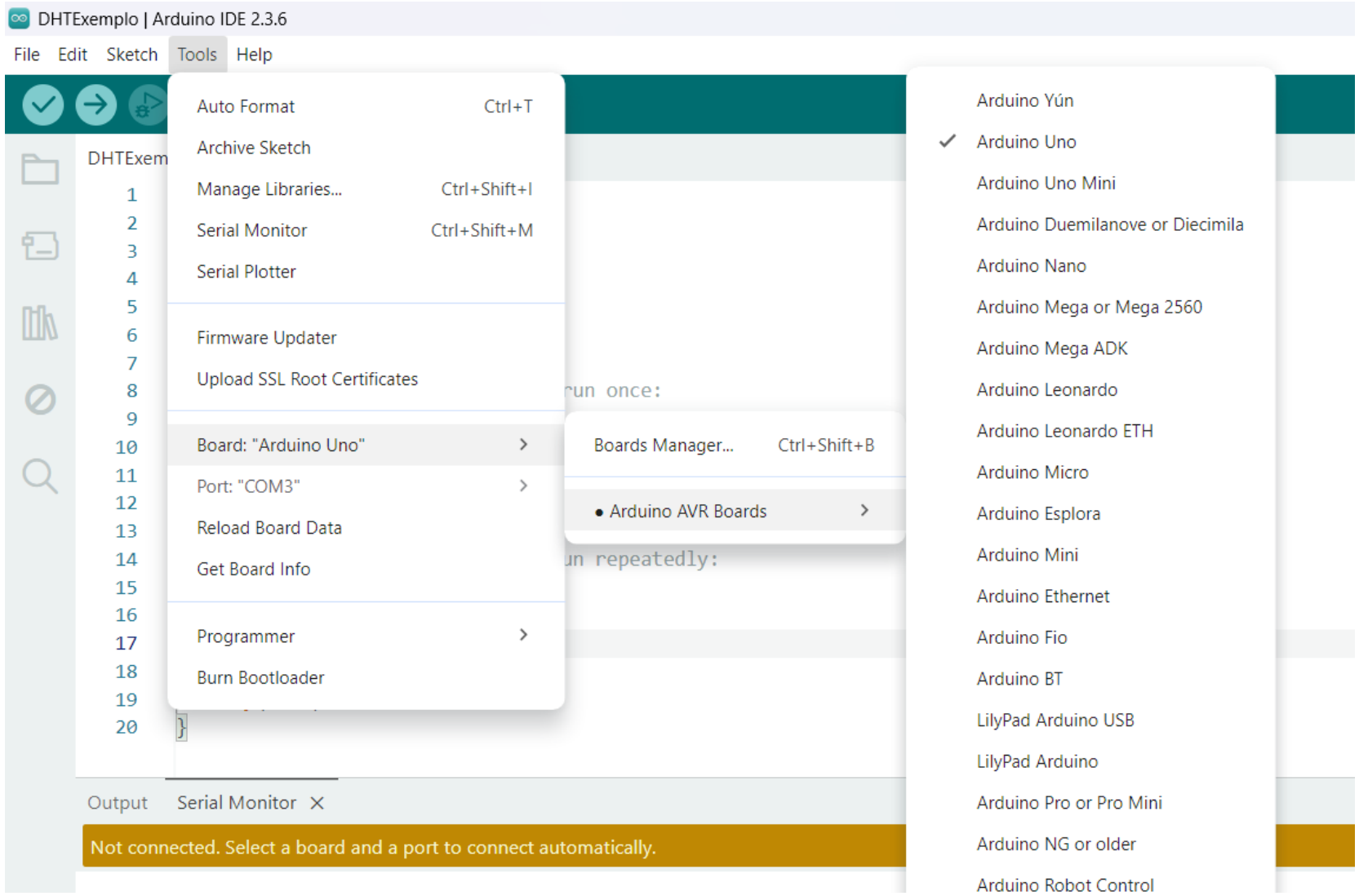




IDE – Ambiente de Programação

https://www.arduino.cc/en/software/

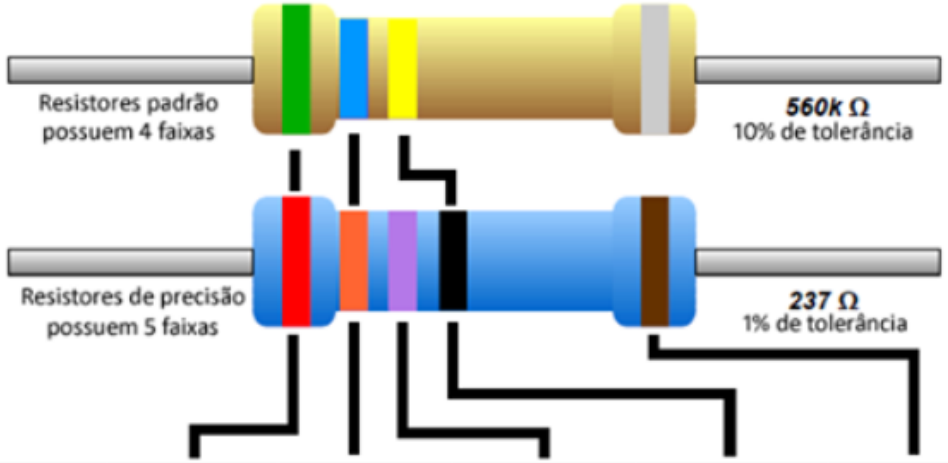
Configuração Placa e Porta



Resistor

Código de Cores

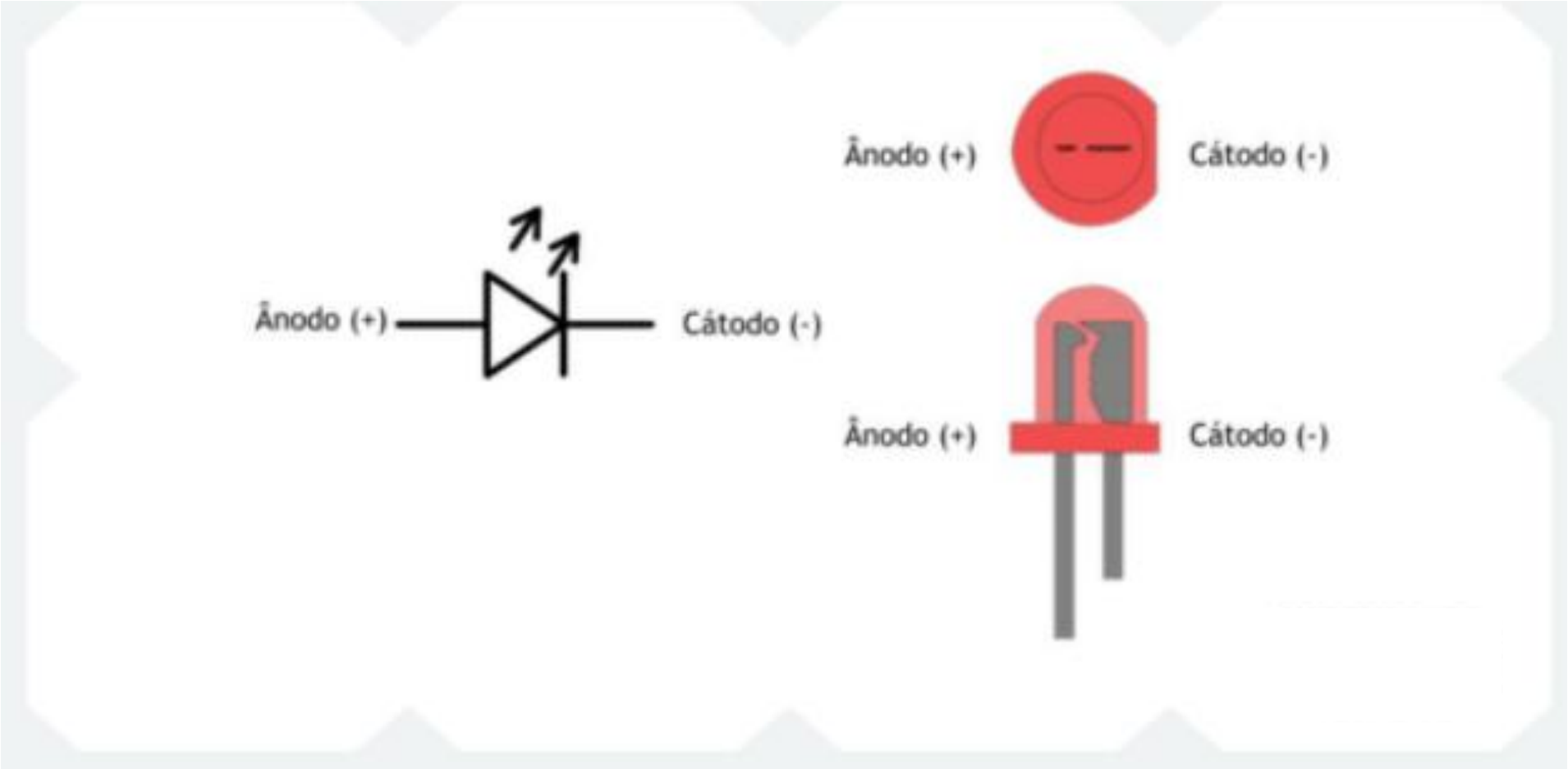
A extremidade com mais faixas deve apontar para a esquerda



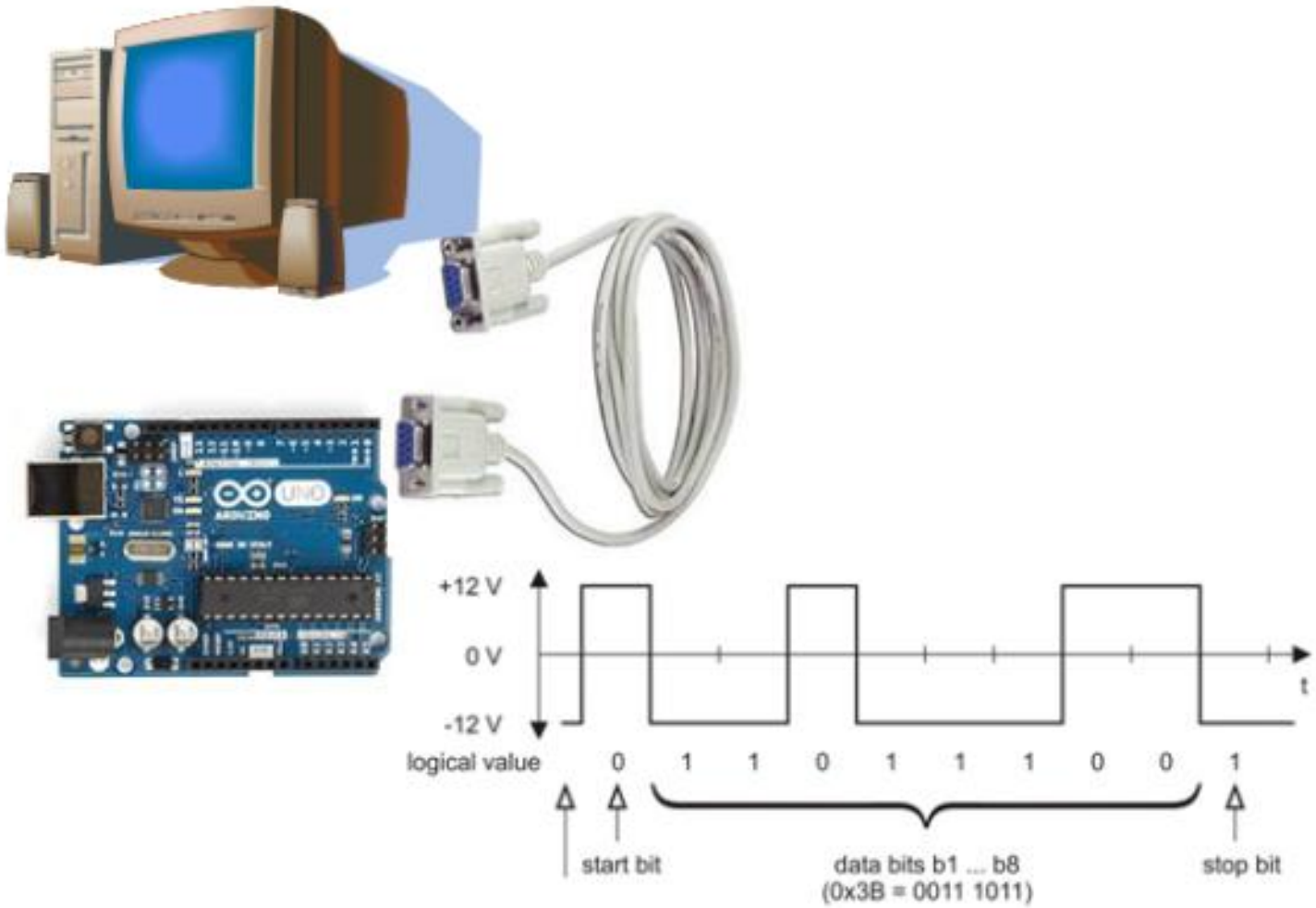
Cor	1ª Faixa	2ª Faixa	3ª Faixa	Multiplicador	Tolerância
Preto	0	0	0	x 1 Ω	
Marrom	1	1	1	x 10 Ω	+/- 1%
Vermelho	2	2	2	x 100 Ω	+/- 2%
Laranja	3	3	3	x 1K Ω	
Amarelo	4	4	4	x 10K Ω	
Verde	5	5	5	x 100K Ω	+/- .5%
Azul	6	6	6	x 1M Ω	+/- .25%
Violeta	7	7	7	x 10M Ω	+/- .1%
Cinza	8	8	8		+/- .05%
Branco	9	9	9		
Dourado				x .1 Ω	+/- 5%
Prateado				x .01 Ω	+/- 10%

LED

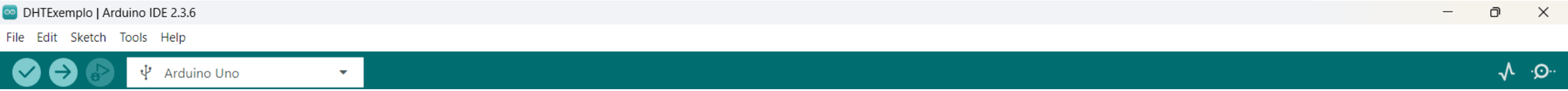
Cuidado com a Polaridade



Comunicação Serial



Primeiro Programa



Compila

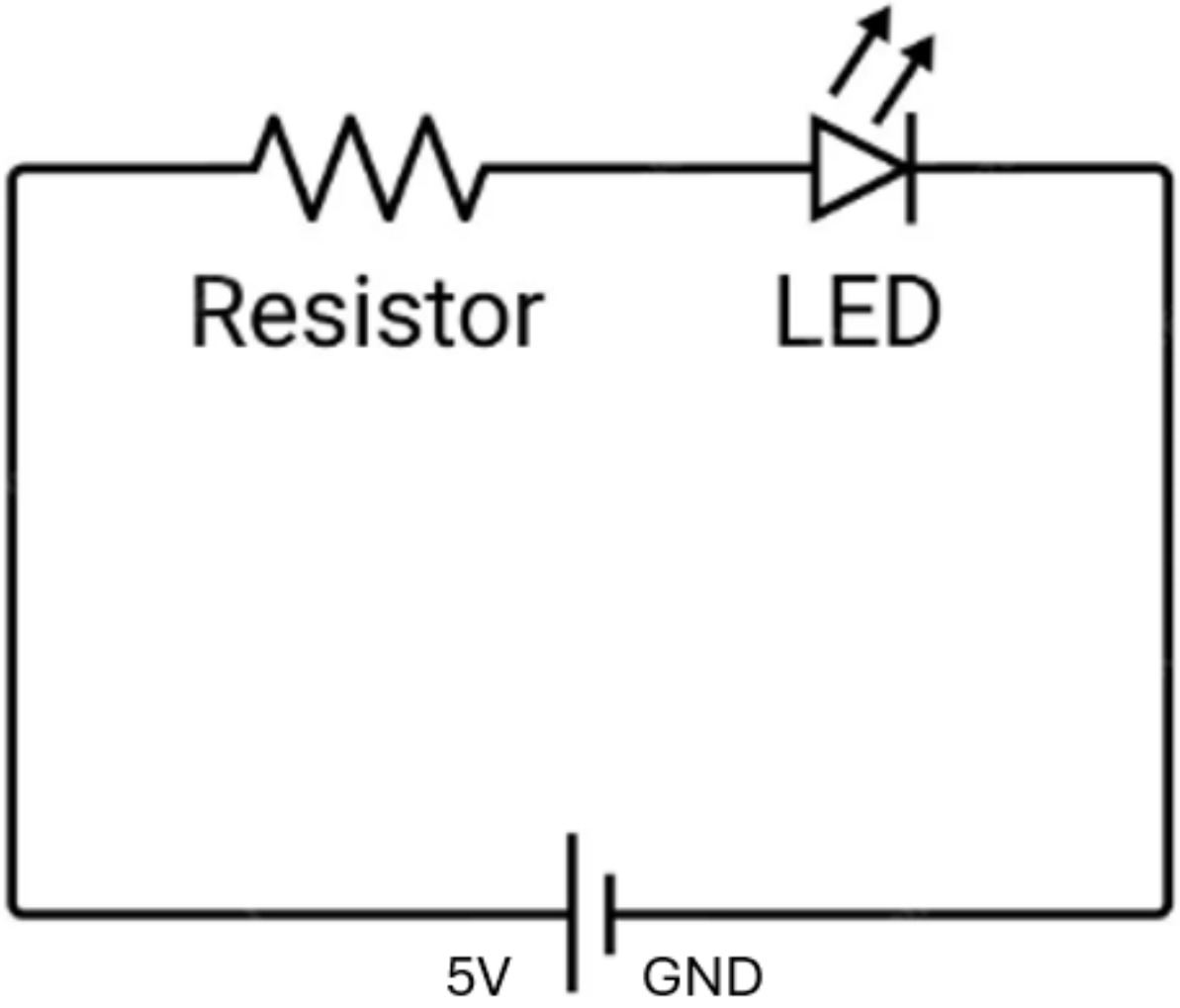
Carrega

Monitor Serial

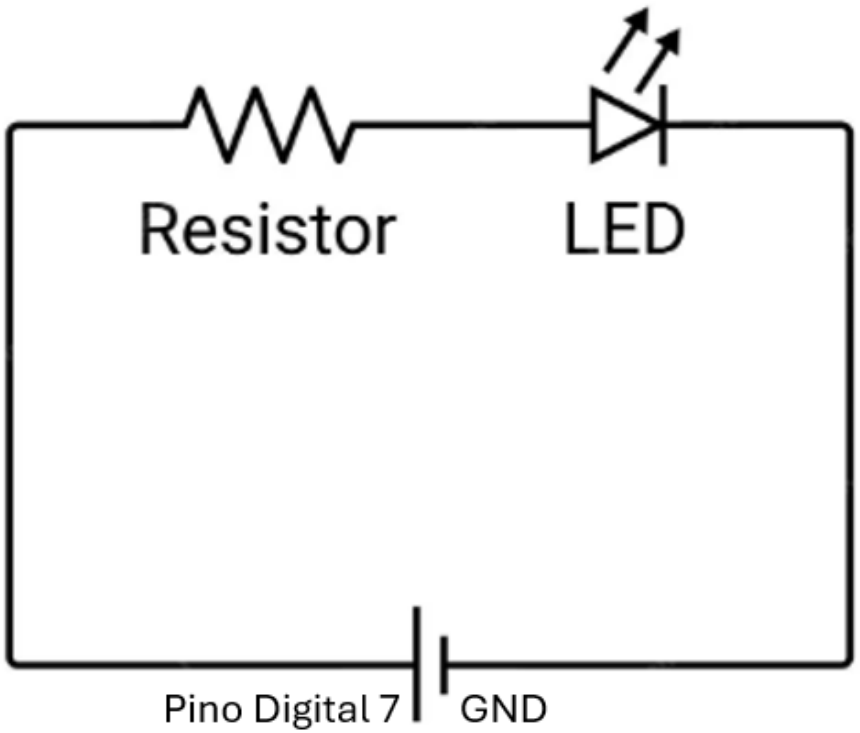
```
void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);
}

void loop() {
  // put your main code here, to run repeatedly:
  Serial.println("Olá");
  delay(2000);
}
```

Acender um LED



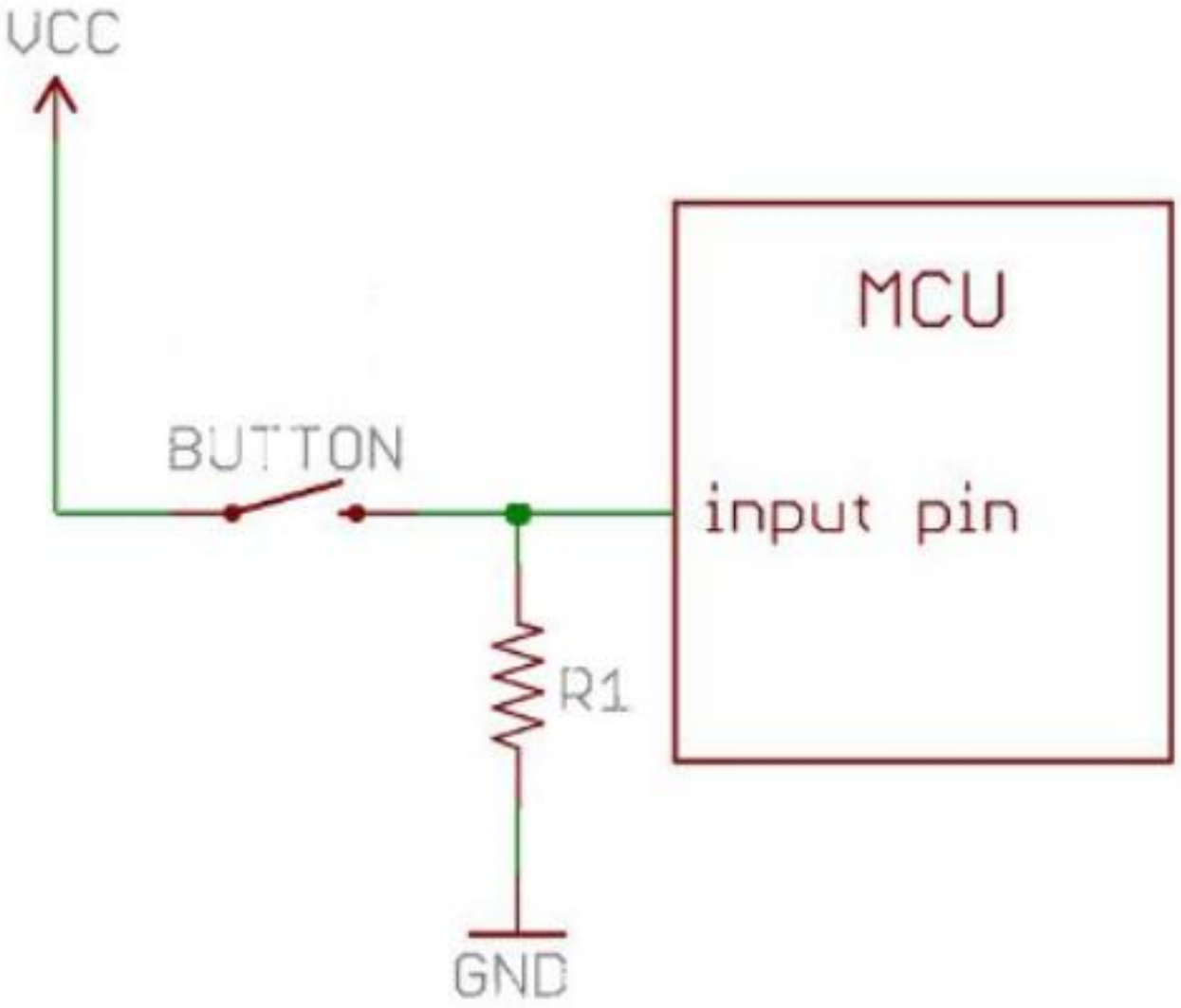
LED Piscar




```
void setup() {
  // put your setup code here, to run once:
  pinMode(7,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
  digitalWrite(7,HIGH);
  delay(1000);
  digitalWrite(7,LOW);
  delay(1000);
}
```

Ler Botão

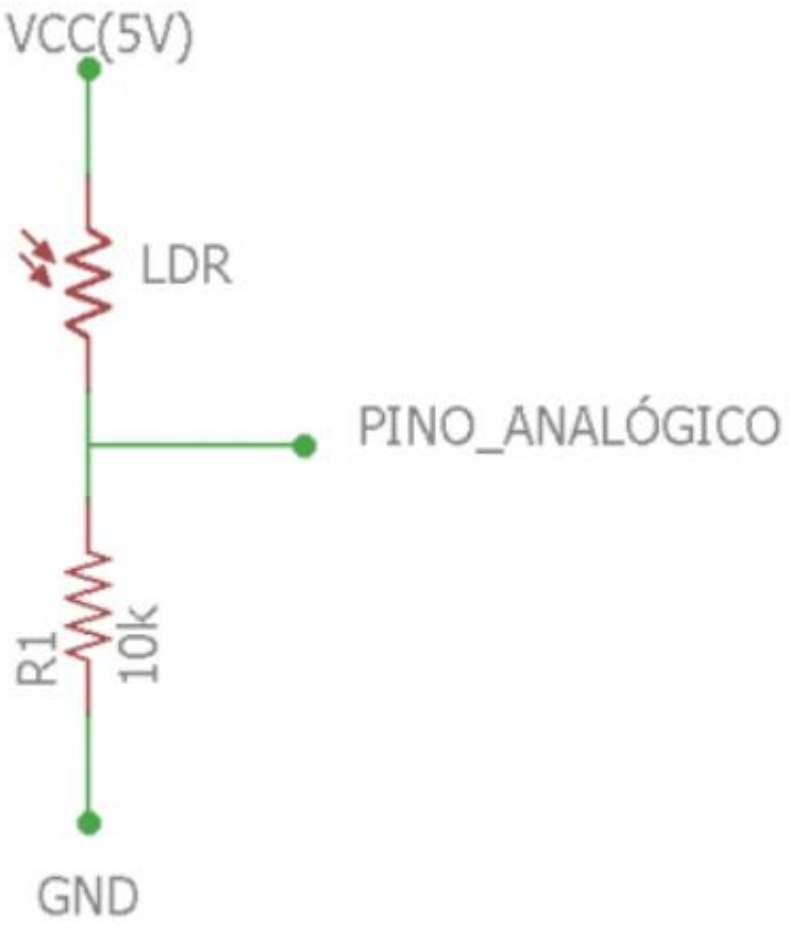


Botão (read) aciona Led (write)

```
void setup() {
  // put your setup code here, to run once:
  pinMode(9,INPUT);
  pinMode(7,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
  if(digitalRead(9) == HIGH)
  {
    digitalWrite(7,HIGH); // Acende LED
  }
  else
  {
    digitalWrite(7,LOW); // Apaga LED
  }
}
```

Sensor analógico – Intensidade Luminosa



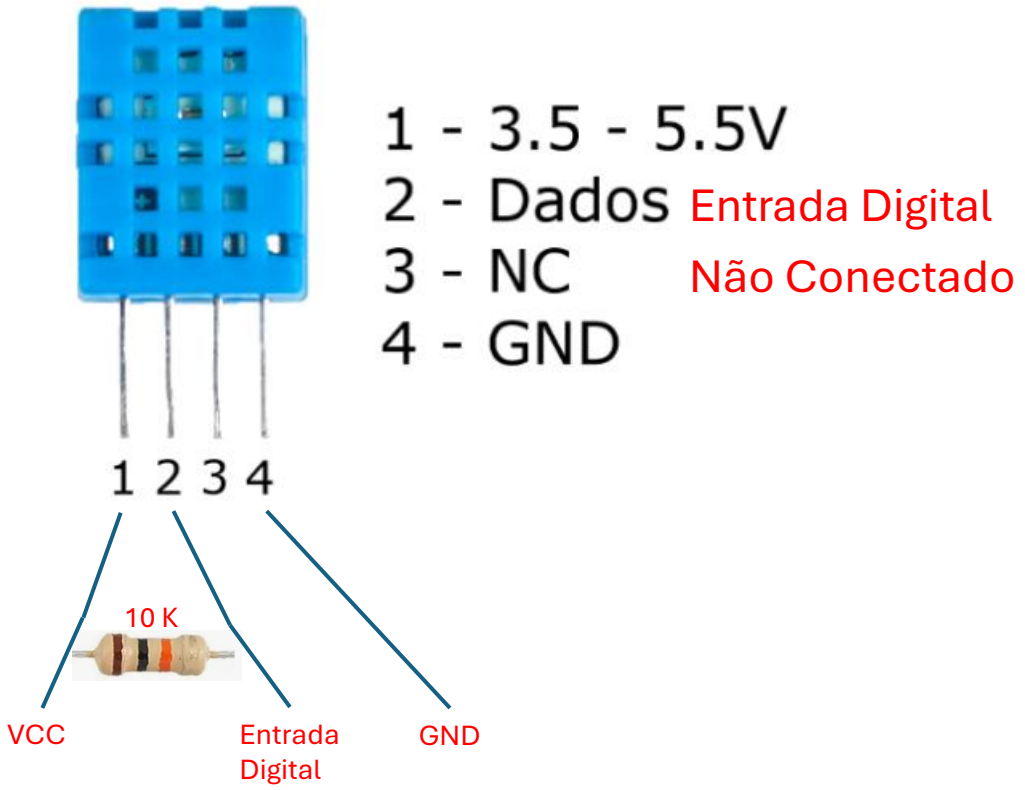

```
void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);
  pinMode(A0,INPUT);
  pinMode(12,OUTPUT);
}

void loop() {
  // put your main code here, to run repeatedly:
  int leitura = analogRead(A0);
  Serial.println(leitura);
  if(leitura > 300){
    digitalWrite(12,HIGH); // Acende LED
  }
  else{
    digitalWrite(12,LOW); // Apaga LED
  }
  delay(2000);
}
```

Sensor de temperatura e umidade

Necessita instalação biblioteca DHT
(próximo slide)

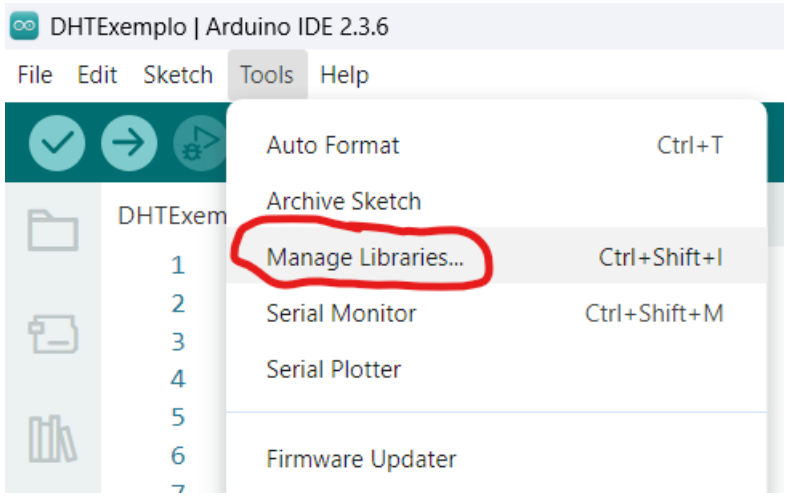
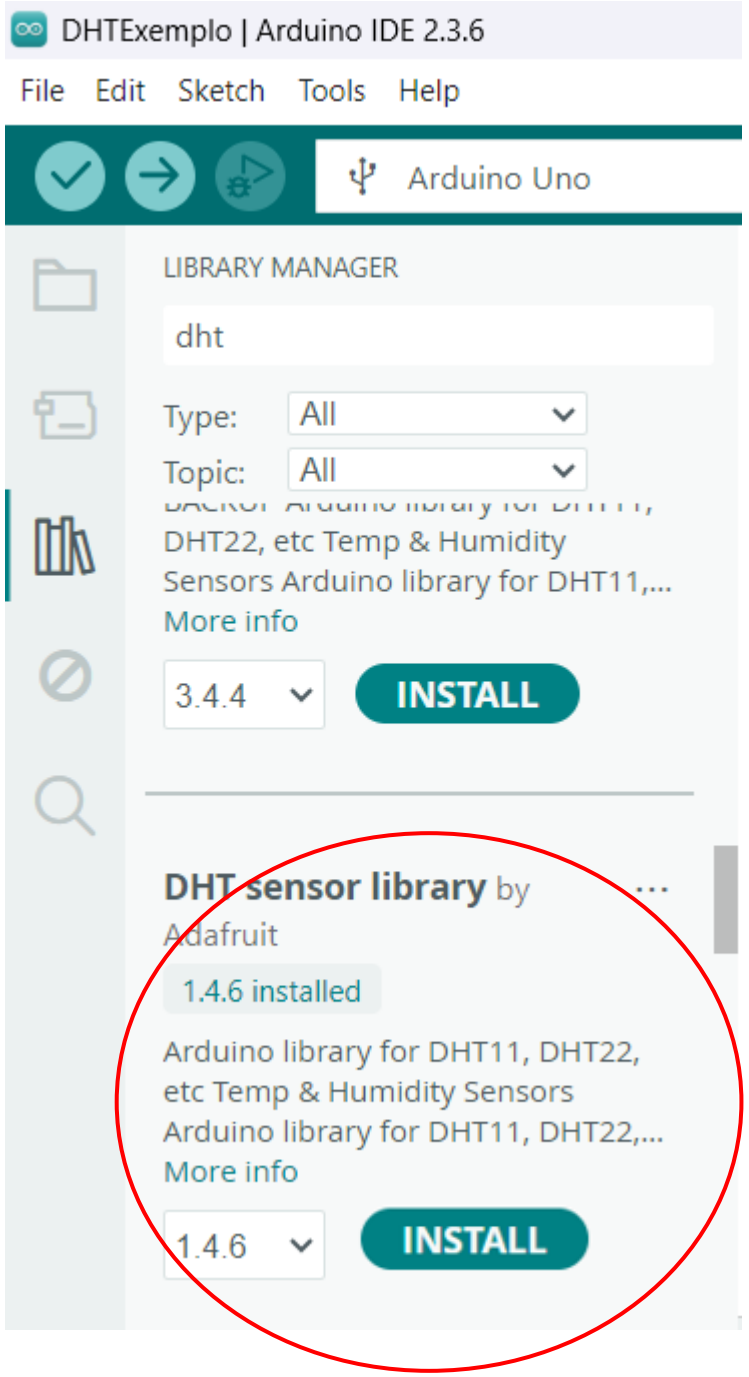
Modelo DHT11




```
#include "DHT.h"

float t = 0;
float u = 0;
DHT dht(9,DHT11); // objeto DHT no pino digital 9

void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);
  dht.begin();
}

void loop() {
  // put your main code here, to run repeatedly:
  t = dht.readTemperature();
  u = dht.readHumidity();
  Serial.println(t);
  Serial.println(u);
  delay(2000);
}
```

Instalação biblioteca DHT